

Bhumika Agarwal

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Software Engineer with **4 years** of experience designing scalable backend systems using **Java, Spring Boot, Microservices, Camunda 7 (BPMN), PostgreSQL, and AWS**. Skilled in modernizing legacy platforms, designing distributed systems, and optimizing API performance, delivering **10K+ dispute cases per month**.

Technical Skills

Languages & Databases: Java, TypeScript, JavaScript, SQL, PostgreSQL, MySQL

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, Microservices, REST APIs, Camunda 7 (BPMN), Kafka

Frontend: React.js, TypeScript, Material-UI, HTML5, CSS3

Cloud, DevOps & Observability: AWS, Docker, Terraform (IaC), Maven, Jenkins, CI/CD, OpenSearch, Splunk

Testing & Concepts: JUnit, Mockito, System Design, Distributed Systems, AI-Assisted Development (Claude, Copilot)

Experience

JP Morgan Chase & Co.

Oct 2024 – Present

Software Engineer II

Mumbai, Maharashtra

- Own the architecture and delivery of the **Overcharge Dispute** capability on an **event-driven platform of 8+ Spring Boot services**, building **Spring Boot + Camunda 7** services that process **10K+ dispute cases/month**.
- Spearheaded the platform's **Apache Kafka** intake pipeline – designing and finalizing the **Avro event schema** – and defined event-publishing payload contracts for the overcharge flow, enabling asynchronous, decoupled processing.
- Modernized legacy **Pega** workflows into **10+ reusable Camunda BPMN components** using **stateless external-task workers**, reducing migration effort by **40%**.
- Automated **AWS** provisioning across **3+ environments** with **Terraform (IaC)** (**60%** faster deployments); built an **OpenSearch** task-indexer and defined **Splunk** field extractions to improve observability.
- Delivered **React + TypeScript** frontend enhancements on a bonds-trading platform, including an **event-tagging feature** for historical event tracking and market-trend analysis, plus backend fixes and improved test coverage.
- Resolved **50+ production issues** across **20+ releases** and established reusable **AI-assisted development** conventions (Claude, GitHub Copilot) to standardize engineering workflows.

Cognizant Technology Solutions

Jul 2022 – Jul 2024

Software Engineer

Mumbai, Maharashtra

- Developed **5+ Spring Boot microservices** and **40+ REST APIs** supporting scalable BFSI transaction platforms.
- Optimized **Hibernate/JPA and SQL** queries across **20+ endpoints**, reducing average **API latency by 30%**.
- Delivered reusable **ReactJS** dashboard components, expanded automated testing with **JUnit 5 and Mockito**, and performed **150+ code reviews**, improving release quality and reducing regression defects.

Projects

Regulatory Approval System | *Java 17, Spring Boot 3, Camunda 7, Spring Security, MySQL, GitHub, Docker*

[Live Demo](#)

- Engineered a BPMN workflow engine using **5 Camunda workers** with **8–48 hr SLA**, retries, orchestration, and failure recovery.
- Implemented **Spring Security, JWT, and 6-role RBAC**, securing **25+ REST APIs** across approval workflows.

UrbanNexus | *Java 17, Spring Boot 3.5, React 19, TypeScript, MySQL, GitHub, Docker*

[Live Demo](#)

- Built a multi-portal platform (Resident, Technician, SuperAdmin) with **3 user roles, JWT authentication, RBAC**, and **25+ REST APIs** on a **React 19 + TypeScript** frontend.
- Designed a **transactional booking engine** with automated allocation, payment orchestration, audit logging, and **JPA lifecycle hooks**.

Education

University of Petroleum and Energy Studies (UPES)

B.Tech. in Computer Science and Engineering

Jun 2018 – Jun 2022

Dehradun, Uttarakhand